An error budget for near-infrared fruit calibration: a reference-replicate design rule and a falsifiable attenuation diagnostic

Panlin Li[a], Yuchang Li[a], Yutong Feng[b,c], Longjie Li[b,c], Ya Liu[d] and Hua Huang[a,e,*]

[a] College of Mathematics and Physics, Xinjiang Agricultural University, Urumqi, 830052, China
[b] College of Mechanical and Electrical Engineering, Xinjiang Agricultural University, Urumqi, 830052, China
[c] Xinjiang Key Laboratory of Intelligent Agricultural Equipment, Urumqi, 830052, China
[d] Urumqi Comprehensive Experimental Station, Xinjiang Academy of Agricultural Sciences, Urumqi, 830013, China
[e] Research Center for Numerical Simulation and Data Analysis, Xinjiang Agricultural University, Urumqi, 830052, China

ARTICLE INFO

Keywords:
measurement uncertainty
error budget
variance component analysis
reference-value uncertainty
near-infrared spectroscopy
calibration validation protocol

ABSTRACT

Near-infrared calibrations for fruit soluble solids content (SSC) or dry matter (DM) are judged by the root-mean-square error of prediction (RMSEP), which holds two variance components that are not the model's. We decompose it into an error budget whose central yield is a design rule for the reference replicates a target R^2 demands, together with a parameter-free, hence falsifiable, attenuation diagnostic. Both presuppose a leakage-free validation protocol, which we specify and audit. In 633 apples with five destructive determinations each, within-fruit variance is 52.30% of the total (intraclass correlation coefficient ICC = 0.477), capping the R^2 of any predictor confined to whole-fruit information at 0.477 for one face and 0.820 for the five-face mean (point estimates); a population bound assuming face-level deviations exchangeable and mean-independent of whole-fruit information, imposed rather than tested; six parameter-free point predictions track measurement. In 4,318 kiwifruit rescanned by 2–5 devices, device and fruit×device terms take 17.07% and 30.33% of spectral variance. Across ten pre-specified operating points ($R^2 = 0.097$–0.820) the measured instrument-related component spans 20.6% of its mean, the fitted trend only 1.1%, while its share of mean squared error rises 5.7-fold, to 24.7%. Random splitting inflates R^2 by 1.41–1.46×. A pre-registered semi-synthetic test reproduced the predicted decay across 13 replicate levels without crossing the ceiling, so the design rule stands as a lower bound, and its 1% empirical margin was directly verified in one of the two blinded targets it changes. Reported performance must be read against a reference-limited ceiling, not unity.

1. Introduction

Non-destructive prediction of internal fruit quality from near-infrared (NIR) or visible–NIR spectroscopy has three decades of practice behind it [1], and its criterion of success is uniform: a destructive chemical determination serves as the reference value, and RMSEP and R^2 on a prediction set are reported. Typical published figures for apple SSC lie at $R^2 > 0.85$ with RMSEP of order 0.5–1.0 °Brix [1, 2, 3], with comparable levels for sugar content in other crops [4]. Portable and imaging instruments [5, 6] and deep-learning models [7, 8] have widened the reach of this paradigm without changing how it is evaluated.

The metric carries a premise that is almost never examined: that the reference value itself is error-free. A destructive determination is also a measurement, with a sampling location, a preparation protocol and an instrument reading, and therefore a variance of its own. Clinical chemistry and metrology have mature frameworks for measurement uncertainty and metrological traceability [9, 10], and the uncertainty of a reference method propagates into comparisons that use it. Errors-in-variables regression and attenuation correction [11, 12] concern error in the explanatory variable, which flattens the slope, whereas the error at issue here sits in the response, leaving the slope alone but inflating the residual variance and lowering R^2. In fruit spectroscopy calibration it is rarely estimated explicitly and still more rarely used to explain the ceiling on reported performance. The consequence is a failure rather than an omission: when the reference value is a single local determination, the reported R^2 cannot tell a good model from a good reference method, and performance reported in this convention loses its standing as a measure of model quality.

Analytical chemistry has recently begun to treat the structure of measurement error itself as an object of study. Ezenarro et al. quantified the covariance and correlation structure of spectral measurement errors across several units of the same miniaturised NIR instrument and derived a correlation index intended to guide preprocessing choices [13]; related work has applied measurement-error analysis to classification with handheld instruments [14] and to a multivariate assessment of the "error quality" of replicate scans [15]. That work defines a contribution type: no new estimator is proposed, the error structure of an existing measurement practice is characterised, and a criterion

*Corresponding author.
huanghua@xjau.edu.cn (H. Huang)
ORCID(s): 0000-0003-0615-9402 (H. Huang)

others can run on their own data is delivered. All of it analyses the spectral side; the counterpart on the reference-value side is the half this paper supplies, since a reference value from destructive sampling likewise has an estimable variance structure, and what that structure defines is a ceiling no model can exceed.

A second neglected component is between-instrument variability. A large body of work has documented degraded cross-instrument performance and proposed correction methods [16, 17, 18, 3], but that work concerns how to repair transfer, not how much of the RMSEP reported on a single instrument was always the part that changes when the instrument changes.

A third issue is not a component of the budget but a precondition for computing it at all. An error budget partitions a reported RMSEP; if that RMSEP was inflated by a leaking validation split, the budget partitions the wrong total. Machine learning has documented how data leakage inflates reported performance [19, 20], chemometrics has issued its own warnings [21], soil NIR calibration has quantified the overfitting caused by not grouping by sampling site [22], and the optimistic bias of cross-validation is pronounced at limited sample sizes [23]. What none of that settles is which grouping level a given dataset requires, a question to be answered layer by layer and audited rather than assumed.

This paper does not propose a better predictive model. It answers a question belonging to analytical chemistry itself: when the reference analytical method carries its own uncertainty, how good can a calibration built on it possibly be? RMSEP is taken apart to ask which parts shrink as the model improves and which are essentially unchanged between the model operating points compared here, using two structurally complementary datasets, one supplying replicate reference determinations and the other replicate scans of the same fruit across instruments. We propose an a priori experimental design tool rather than an after-the-fact critique. Its core is the design rule of §3.5: once the variance ratio σ_f^2/σ_a^2 has been estimated on a small subset of the target batch, inverting it gives the minimum number of independent reference determinations per sample required for a target R^2 to be physically attainable, the attainable ceiling itself (§3.1) being a worked example rather than the end point. It is accompanied by a second tool, an attenuation-ratio diagnostic that any laboratory holding per-face (or per-point) replicate reference determinations can run on its own data. Both presuppose a validation protocol that does not leak, which is why the individual and the acquisition-day grouping level are specified here as parallel candidates, each audited for necessity.

Methodologically we contribute a form rather than an estimator: the variance-component model is required to deliver point predictions that contain no free parameters, so that the decomposition can be refuted rather than merely described, and those predictions are confronted with measurement, including a semi-synthetic experiment designed so that they could fail. Variance components are long established in chemometrics; inverting a reliability into a required number of observations is the Spearman–Brown formula [24, 25], and asking how that reliability responds when observations on one facet are increased is what decision studies in generalizability theory do [26]. Neither a new regressor nor a new preprocessing method is proposed; what is new is the destination and the form, and it is the form that separates this work from the existing "report the measurement uncertainty" genre. The intent is to give calibration design and performance assessment in non-destructive fruit quality testing a basis that can be computed in advance and rechecked on the reader's own data.

2. Materials and methods

2.1. Datasets

Two datasets carry the error budget (Table 1) in a relation of division of labour, not mutual corroboration: the per-face replicate structure unique to the apple data estimates the reference-side component, the cross-device replicate scans unique to the kiwifruit data the measurement-side component. Public datasets are typically "one fruit, one spectrum, one label" and structurally cannot yield the former.

A third dataset appears once and is declared here so that the count is not misread: the public DeepHS-Fruit time-series hyperspectral set of post-harvest ripening sequences (five species, 3,405 VIS spectra) [27], used in §3.4 and the Supplementary Material. It contributes nothing to either variance component and to no claim the paper makes; it is the dataset on which we initially read a day-level leakage effect and then withdrew that reading, kept because the trap it illustrates is worth reporting.

2.2. Variance-component model

For the per-face reference values a one-way random-effects model is used

$$
\begin{aligned}
&Y_{ij} = \mu + a_i + e_{ij}, \\
&a_i \sim (0, \sigma_a^2), \quad e_{ij} \sim (0, \sigma_f^2), \quad j = 1, \dots, 5,
\end{aligned}
\tag{1}
$$

where a_i is the whole-fruit deviation of fruit i and e_{ij} the deviation of its j-th face from that fruit's mean. The design is balanced, so the classical ANOVA moment estimators $\hat{\sigma}_f^2 = \mathrm{MS}_{\mathrm{within}}$ and $\hat{\sigma}_a^2 = (\mathrm{MS}_{\mathrm{between}} - \mathrm{MS}_{\mathrm{within}})/5$ are used rather than REML, which keeps the decomposition directly recheckable from the mean squares [30] but still relies on the additivity and homoscedasticity assumptions of the model. The ICC, the minimal detectable difference derived from it and

Table 1
Data card. The two datasets carry different component-estimation tasks in this paper.

	Apple (reference-side component)	Kiwifruit (measurement-side component)
Source	Collected by Xinjiang Agricultural University in 2025; previously unpublished; not publicly released under the holding institution's data policy, available from the corresponding author on request	Public, on figshare under CC BY 4.0 [28], first described by Wohlers et al. [29]; used exactly as deposited, with no augmentation from that work
Origin / batch	Merchant-declared origins Gansu, Shandong and Xinjiang; all three groups purchased in one 2025 acquisition from a single wholesale market in Xinjiang	Origin not distinguished
Sample size	633 fruit $\times$ 5 faces $=$ 3,165 reference determinations after cleaning; 3,149 rows remain after fruit–face pairing (16 faces have no matching spectrum). The variance-component analysis uses the former (fully balanced), spectral modelling the latter	5,418 fruit / 11,982 spectra
Key replicate structure	Five faces per fruit: top, bottom and three equatorial side faces 120° apart	Cross-device replicate scans: 4,318 fruit scanned by 2–5 devices (only 90 by all five); the further 1,100 scanned once contribute nothing to the instrument component
Reference determination	After destructive sampling, a digital refractometer gives one reading per face (nominal accuracy ± 0.2 °Brix)	SSC and dry matter (DM)
Spectra	Reflectance, 346 bands over 391–1001 nm	222 bands over 402–1065 nm, the common usable range of the five devices
Cleaning rule (fixed before analysis)	Remove whole fruit containing any face with SSC $>$ 20 °Brix (the physiological ceiling is 18–20, so higher values are entry errors); keep values <5 °Brix, since unripe or rotten tissue genuinely reaches 3–5 and removing them would shrink the finding a priori	Drop rows with missing values and the wholly missing 1068–1137 nm columns, absent on all five devices, leaving 222 bands; a later missing-channel mask removes 0 of those 222. The instrument-side analysis uses only the subset scanned by ≥ 2 devices
Use in this paper	Within-fruit reference variability, attainable ceiling, $1/m$ design rule, attenuation diagnostic	Between-instrument error budget, comparison of two model operating points
SHA-256 (first 16)	`b5e923b4d05be65f` (cleaned analysis table; the two raw workbooks are `e77fb341681ac5e7` and `bfdbeba89ef79b80`)	`37157f2c42a5d441` (original deposit as downloaded)

agreement analysis in method comparison are established measurement tools, as are the known limitations of judging agreement by a correlation coefficient [31, 32, 33]. Intervals for the variance components and the ICC come from bootstrap resampling of fruit, the five faces moving together ($B = 2000$).

If the reference value is the mean of m faces, $\bar{Y}_i^{(m)}$, then any model that can perceive only whole-fruit information (i.e. can predict only $\mu + a_i$) satisfies

$$R^2 \leq \frac{\sigma_a^2}{\sigma_a^2 + \sigma_f^2/m}, \qquad \mathrm{RMSE} \geq \frac{\sigma_f}{\sqrt{m}}. \tag{2}$$

Equation (2) constrains the population R^2, whereas the R^2 reported here is a finite-sample score referenced to the training-fold mean and so never lower than under the test-mean convention (§2.3), which leaves it not guaranteed to obey the bound exactly in finite samples. Every comparison drawn here between a measured score and this ceiling, and equally the six point predictions of §3.2 and the design rule of §3.5, is approximate to the 0.001 order quantified in Supplementary S2, and the bound is not claimed as an exact property of the reported statistic.

Two conditions are required, and it is worth separating them. The first is exchangeability of the five faces: the deviations e_{ij} are mutually uncorrelated within a fruit, so the variance of their mean is exactly σ_f^2/m. The second is that the face deviations carry no information the predictor can see. Zero covariance, $\mathrm{Cov}(a_i, \bar{e}_{i\cdot}) = 0$, does not suffice, because it constrains only linear dependence whereas the cross-term that must vanish when a predictor g is subtracted is $\mathbb{E}[(a_i - g)\,\bar{e}_{i\cdot}]$, which for nonlinear g a covariance does not control. The condition stated here is the stronger one: the mean deviation $\bar{e}_i^{(m)}$ over the m faces serving as the reference value is mean-independent of every whole-fruit quantity a predictor can condition on, $\mathbb{E}[\bar{e}_i^{(m)} \mid a_i, S_i] = 0$, with S_i the spectrum, and it is required separately for each m in use. Three explicit constructions in Supplementary S1 show why each strengthening is needed and that under zero covariance alone Eq. (2) holds only against predictors affine in the whole-fruit information. Both conditions are assumptions rather than consequences of writing the model down. The first is tested in §3.2; the second is not testable on this dataset in either version, because it would need technical replicates on the same face (§4.3), and it is imposed. Under merely approximate mean independence the expression is an approximation whose error is of the order of the residual predictable part of $\bar{e}_i^{(m)}$.

The m "replicates" here are destructive samplings at m different positions on the same fruit, not m replicate analyses of the same sample, so σ_f^2 is the spatial-subsample variability defined by the five-point scheme, pooling between-face chemical heterogeneity, sampling and preparation variability, and refractometer reading error, which are inseparable here. Equation (4) accordingly asks how many approximately independent spatial subsamples per fruit are needed, not how many parallel determinations the reference method must perform; Supplementary S1 sets out the distinction and records that the five faces were the pre-existing scheme of this collection, so the $1/m$ test of §3.2 is an in-sample check.

2.3. Modelling and validation

Unless stated otherwise, partial least squares regression (PLS) is used, with the number of components selected by grouped cross-validation within the training set. Using one regressor is deliberate. Eq. (2) bounds every predictor that perceives only whole-fruit information and Eq. (4) inherits that property, so neither belongs to the regressor: a stronger learner can carry a study closer to the ceiling but cannot raise it, and a model comparison would vary the distance to the bound while leaving the bound itself, the object measured here, untouched. PLS fills the role because it is the reference method of spectroscopic calibration [34, 7] and its latent structure reduces tuning to one decision, the component count, taken by grouped cross-validation inside each training fold. Where the restriction costs something it is stated: other model classes and parameter spaces were not individually excluded, so the cause of the low apple-side performance remains a candidate rather than a demonstration (§4.1). The "weak model" operating point of §3.3 is a deliberate exception, its component count fixed at 10 to construct a poorly performing comparison point. Preprocessing compares raw, SNV, Savitzky–Golay first derivative and their combinations [34, 7], following published best-practice guidance [35] and comparable quantitative applications [36]: scatter correction and derivative filtering address the two artefacts that dominate diffuse reflectance on intact fruit, multiplicative path-length variation and baseline drift, and are compared on the data rather than assumed.

The total sum of squares in R^2 is referenced to the training-set mean rather than the test set's own mean, because the training mean is the only baseline available at prediction time. An algebraic identity makes the training-mean denominator never smaller and the resulting R^2 never lower; replaying every split used here without refitting, the two conventions differ by at most 0.001 in R^2, below the precision at which it is reported. The identity and the measured denominator ratio are in Supplementary S2.

Except for the row-wise random split used deliberately as a leakage control in §3.4, and for the acquisition-day grouping described next, every performance estimate is grouped by individual: multiple spectra of the same fruit never straddle the train/test boundary. For the time-series spectra, grouping by acquisition day is evaluated as a second grouping level parallel to the individual one, not a further layer on top of it: it holds out whole acquisition days and does not block the individual level, so spectra of the same fruit taken on different days do fall on both sides of the boundary. The two levels are audited separately, and neither estimate is a generalization estimate for unseen fruits and unseen acquisition days at once.

Replication and uncertainty reporting. Everything depending on a random split runs in two stages: feasibility is verified on a single development seed (Pilot), then the whole analysis is rerun on five seeds fixed in advance (Formal), and every split-dependent quantity in the text is the Formal result. A second class depends on no random split, namely the variance components and ICC, the attainable ceiling, the label-only baseline and the spectral-quality and saturation diagnostics; these are seed-independent, so the original computation was retained. Intervals for split-dependent quantities use a two-level cluster bootstrap with seeds as clusters, because repeats under one seed share a single random stream and an i.i.d. bootstrap underestimates interval width; the reported SDs reflect only splitting and modelling variance, not hardware jitter. Every ratio-valued quantity is computed as a ratio of means rather than a mean of ratios, because predictions such as Eq. (3) are themselves ratios of expectations. The seed list, the repeat count of every analysis, the bootstrap construction and the measured gap between the two ratio conventions are in Supplementary S2.

Computation runs on CPU. Given the seed, a rerun on the same architecture and BLAS implementation returns the reported values bit for bit; across architectures it does not, because the inner cross-validation MSE curve is nearly flat in weak-signal configurations, so floating-point differences can move the component selection. A full replay on a second architecture, reported in Supplementary S2, leaves most checked quantities unchanged, moves only weak-signal ones and reverses no statistical conclusion. The dependence is removable rather than merely disclosed, since the component count selected for every configuration is released with the model records and holding it fixed reproduced the reported value to the sixth decimal place on the other architecture. The software environment that produced the reported numbers was not recorded at the time; the pinned environment released with the code is the one in which the replay was run, and is labelled as such.

2.4. Pre-registration and traceability

The statistical analysis plan, the cleaning rules and every criterion were archived before the final results were seen. Every variance component, R^2, RMSE, ratio and confidence interval reported here is produced by a script and written to a workbook, and each such number in the figures is traceable to a specific table; dataset metadata are traceable to the data card instead. Quantities the text reports but derives from other stored components are themselves exported, so that "traceable" means stored, not merely recomputable. Process-level intermediates such as the bootstrap resamples can be replayed from the scripts and seeds. Any deviation from the analysis plan is entered in the protocol's deviation log.

3. Results

3.1. Within-fruit variability of the reference value and the resulting ceiling

After cleaning, $n = 633$ fruit, $\hat{\sigma}_a^2 = 2.327$ and $\hat{\sigma}_f^2 = 2.552$: the within-fruit (between-face) variance is 52.30% of the total (ICC = 0.477, bootstrap 95% CI [0.436, 0.515]). The within-fruit SD is $\hat{\sigma}_f = 1.597$ °Brix and the standard error of the five-face mean $\hat{\sigma}_f/\sqrt{5} = 0.714$ °Brix; the SD of the single-face values is larger, 2.207 °Brix, because it also carries $\hat{\sigma}_a$, and the two must not be confused. The median within-fruit range is 2.80 °Brix, 44.1% of fruit exceed 3 °Brix and 20.7% exceed 5 °Brix (Fig. 1a); the median within-fruit SD across five faces, 1.110 °Brix, is 5.6 times the refractometer's nominal accuracy.

The proportion differs markedly between the three origin groups: 38.5% for Shandong, 52.5% for Xinjiang and 66.5% for Gansu (Fig. 1b), all bought in the same acquisition from the same wholesale market (§2.1). No hypothesis test of the between-group differences is performed; these are point estimates only. The corresponding attainable R^2 ceilings run from 0.615 down to 0.335, nearly a factor of two, so the ceiling is not a species constant and must be reported per collection.

By Eq. (2), a single face as the reference value gives an R^2 ceiling of 0.477 and the mean of five faces 0.820 (Fig. 1c), both point estimates quoted as such throughout. Propagating the ICC interval gives [0.436, 0.515] for a single face and [0.794, 0.842] for the five-face mean; the conservative reading, under which a model is hardest to declare ceiling-bound, takes the upper limits 0.515 and 0.842, and every statement about this ceiling should be read as carrying that interval.

An immediate objection is that the within-fruit variability merely restates the known calyx–stem gradient. The gradient is confirmed, the bottom exceeding the top by 0.405 °Brix (paired $t = 4.42$), but it is only 36% of the median within-fruit SD and the fixed effect of face position explains just 0.95% of the within-fruit variability. Restricting to the three side faces, which lie on the same equatorial band and carry no stem–calyx gradient, still leaves a between-face variance of 2.4396, i.e. 95.6% of the value 2.5515 estimated from all five faces. The variability therefore has almost no structure in the spatial variables this dataset records, which is not to say it has none, since it may relate to unrecorded factors such as sun exposure or canopy position; for that reason standardising the sampling position alone will not eliminate it. Spatial dynamics of sugars and cell state within apple flesh have direct cytological evidence in watercored fruit [37], a physiological disorder, so that evidence is compatible with the dispersion seen here but is not extended to sound fruit, and with no technical replicates on the same face (§4.3) this dataset cannot attribute the dispersion to biological heterogeneity either. The full control is in Supplementary S5.

3.2. Six point predictions: theory versus measurement

Equation (2) and its corollaries give four groups totalling six parameter-free point predictions checkable against the data (Fig. 2; all values are in the companion workbook).

The first group is the $1/m$ scaling. Taking the mean of the first m sampling faces of each fruit in a fixed order, the measured variance deviates from the theoretical value $\sigma_a^2 + \sigma_f^2/m$ by -1.61% ($m = 2$), $+2.30\%$ ($m = 3$) and $+0.72\%$ ($m = 4$). Both variance components are estimated once from the complete $m = 5$ design, so the test is not circular; a fixed subset is used rather than the average over all $\binom{5}{m}$ combinations, so sensitivity to subset choice is untested, and this remains an in-sample extrapolation.

The second group tests exchangeability across faces. The mean off-diagonal element of the covariance matrix is 2.331, consistent with the value $\hat{\sigma}_a^2 = 2.327$ predicted under exchangeability, and a permutation test of compound symmetry cannot be rejected ($p = 0.261$). Failure to reject is not proof that the scaling holds exactly.

The third group is the label-only baseline. Predicting the SSC of one face from the mean of the other four faces of the same fruit has theoretical value $1-(\sigma_f^2+\sigma_f^2/4)/(\sigma_a^2+\sigma_f^2) = 0.3462$ against a measured 0.3456. This expression contains no free parameter, is the sharpest of the group, and involves no train/test split, so the 0.001-order note of §2.2 does not apply to it.

The fourth group is the attenuation ratio, the only one carrying a confidence interval, treated separately below.

Suppose a model perceives only whole-fruit information and extracts the same signal under both targets, so that the variance it explains is the same for both; the ratio of R^2 between the targets then equals the ratio of their variances. The second half of that supposition is an assumption in its own right, since two predictors can both be confined to whole-fruit information and still fit different amounts of it; the paired design below makes it plausible and the interval test would expose it if it failed. One subtlety decides the ratio: the "whole-fruit" target is the mean of five faces, so it still carries noise

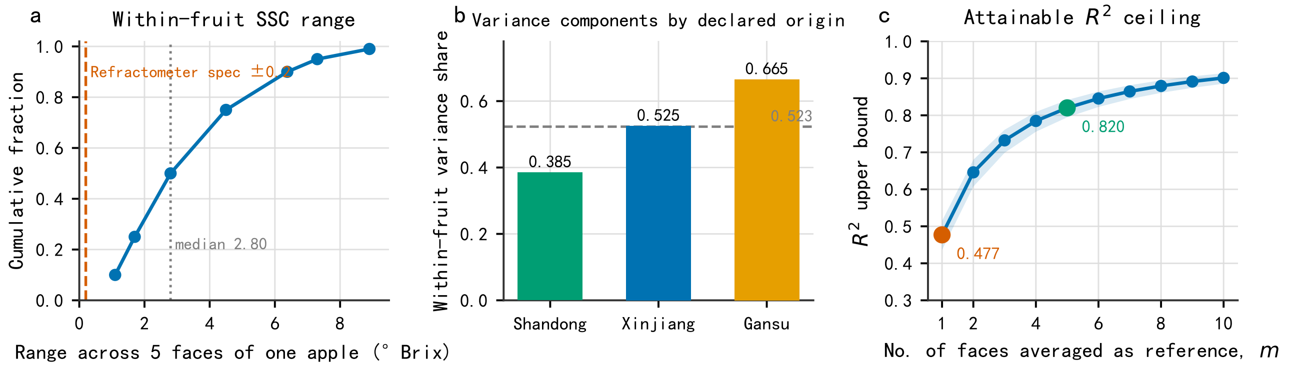

Figure 1: Within-fruit variability of the reference value. (a) Cumulative distribution of the SSC range across the five faces of a fruit; dashed lines mark the refractometer's nominal accuracy and the median range. (b) Within-fruit share of the total variance by origin group; the horizontal dashed line is the pooled value 0.523. (c) Attainable R^2 ceiling against the number of faces m averaged into the reference value; the shaded band is the bootstrap 95% interval of the ICC.

$\sigma_f^2/5$, and the prediction is not the ICC but

$$\frac{R^2_{\text{face-level}}}{R^2_{\text{whole-fruit mean}}} = \frac{\sigma_a^2 + \sigma_f^2/5}{\sigma_a^2 + \sigma_f^2} = 0.5816. \qquad (3)$$

It is tested under a paired design in which the two targets share the same splits, the numerator and denominator being the raw-preprocessing face-level and fruit-mean R^2 values of the main apple table; the corresponding figures in the leakage controls of §3.4 come from a separate paired run with its own splits and can neither be interchanged with these nor used to recompute the ratio. The measured ratio is 0.564, with a two-level seed-clustered bootstrap 95% confidence interval [0.531, 0.596] that contains the prediction 0.5816. The comparison is discriminating rather than merely permissive: the ICC itself (0.477, which corresponds to a noise-free whole-fruit target, i.e. the limit $m \to \infty$) falls outside that interval, so the data distinguish between two candidate predictions differing by only 0.10 and select the one the variance-component model implies. Both numerator and denominator are R^2 values of magnitude only 0.06–0.12, so the ± 0.03 width is driven mainly by the weakness of the spectral signal rather than by unstable splitting, the margin by which the interval clears the ICC is correspondingly modest, and nonparametric bootstrap intervals have their lowest power in exactly this small-effect regime [38]. This discrimination is therefore the weakest of the point predictions and should be rechecked where the spectral signal is stronger.

3.3. Between-instrument variability and its weak dependence on model quality

In the kiwifruit data there are no technical replicates within a device, so the instrument×sample interaction and the scan/repositioning noise are inseparable and only their sum can be reported. On that basis the variance of the SNV-corrected spectra decomposes into 52.60% between fruit, 17.07% device main effect and 30.33% fruit×device pooled residual (Fig. 3a): nearly half of the spectral variation originates in the measurement system rather than in biological difference. The device main effect is a systematic bias, identifiable under a crossed design and in principle correctable, which is precisely the target of calibration-transfer methods [16, 18, 3].

The predictive-level result is more direct. At two model operating points, a weak model (SNV, full spectrum, 10 components) gives $R^2 = 0.393$ and RMSEP $= 1.809$ %DM and a strong model (first derivative, 700–1065 nm, cross-validated component count) $R^2 = 0.819$ and RMSEP $= 0.988$ %DM. Their RMSEPs differ by a factor of 1.83, but the part of RMSEP corresponding to the dispersion of predictions for the same fruit across devices barely moves: 0.495 versus 0.490 %DM. Two points cannot distinguish invariance from a shallow trend, so the operating range was scanned. The scan uses a different fold rule, assigning fruit at random to five folds under each fixed seed rather than the deterministic grouped k-fold used above, because a slope needs an uncertainty; its numbers are therefore its own and are not interchangeable with the two above. Ten configurations crossing five preprocessing methods, three spectral ranges and five component budgets were fixed in advance, the two operating points above being two of them; all ten are listed in Table S1 of Supplementary S3. They span R^2 from 0.097 to 0.820, over which the total RMSEP falls by 55% (2.207 to 0.986 %DM).

Across that scan the pooled component stays between 0.457 and 0.562 %DM (mean 0.505), while its share of mean squared prediction error rises from 4.3% to 24.7%, a factor of 5.7 (Fig. 3b). Regressing the component on R^2 gives a slope of $+0.0073$ %DM per unit R^2, with a seed-clustered bootstrap 95% interval of [+0.0038, +0.0109] that excludes zero, so the component is not strictly invariant and that is reported

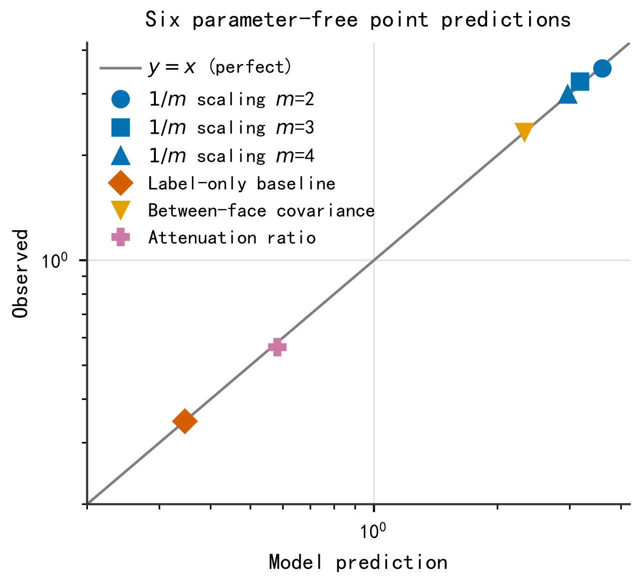

Figure 2: Parameter-free point predictions: abscissa, the value predicted by the variance-component model; ordinate, the measured value; grey line, $y = x$.

rather than invariance claimed. The size of the trend is what matters. The regression accounts for 0.5% of the variance of the component across configurations, and over the entire R^2 range the fitted trend predicts a change of +0.005 %DM, i.e. 1.1% of the component's own magnitude, while the denominator it is divided by shrinks by more than half. That 1.1% is the trend, not the observed movement: the measured values span 20.6% of their mean across the ten configurations, and the residual SD of ±0.03 %DM shows that almost all of that spread is configuration-to-configuration scatter unrelated to model quality. All five fitted diagnostics are exported alongside the slope.

Two qualifications stand: the quantity is the pooled device×sample interaction and scan/repositioning noise rather than pure instrument repeatability, and the configurations differ simultaneously in preprocessing, spectral range and component budget, so the scan is a sweep of operating points rather than a controlled ablation. Within those qualifications, model improvement compresses the compressible part of the error and leaves the pooled component very nearly untouched, so the share of the latter rises steeply.

3.4. Grouping levels of the validation protocol

The components above are meaningful only if the validation protocol itself is leakage-free. Cross-validation practice and nested feature selection have been discussed systematically in the methodological literature [39, 40]; what this section adds is that the grouping level itself must be determined layer by layer from how the data were generated. It reports the one leakage confirmed here; a second, day-level reading withdrawn after a control test is documented in full in the Supplementary S4, because the trap it illustrates is easy to miss.

In the apple data, replacing a row-wise random split by grouping by fruit lowers R^2 from 0.0878 to 0.0625 (face-level SSC) and from 0.1692 to 0.1195 (whole-fruit mean SSC), inflation factors of 1.41 and 1.42; under first-derivative preprocessing the same comparison gives 0.0917 → 0.0631 and 0.1580 → 0.1079, factors of 1.45 and 1.46, so across the four combinations the inflation is 1.41–1.46×. The seed-clustered bootstrap 95% confidence intervals are [1.21, 1.71], [1.27, 1.63], [1.25, 1.67] and [1.30, 1.71], all four lower limits sitting clearly above 1 (Fig. S2 of the Supplementary Material). The mechanism is that the remaining faces of the same fruit fall into the training set and leak whole-fruit information.

On the time-series hyperspectral data this work initially read a second, deeper layer of leakage: predicting the imaging day index of a single spectrum gives $R^2 = 0.887$ when grouped by fruit and 0.705 when grouped by acquisition day. That reading does not survive a control test and we withdraw it. The day index is numbered from the acquisition-day label, so holding out a whole day also removes an entire level of the target; rerunning with destructive measurements not derived from the grouping variable leaves only 2 of 6 combinations above the pre-specified threshold, both on the one target still strongly correlated with acquisition day. We therefore do not claim a day-level mechanism independent of individual-level leakage. The episode is nonetheless a reproducible protocol trap: when the target is derived from the grouping variable, grouping by it manufactures a leakage-like drop out

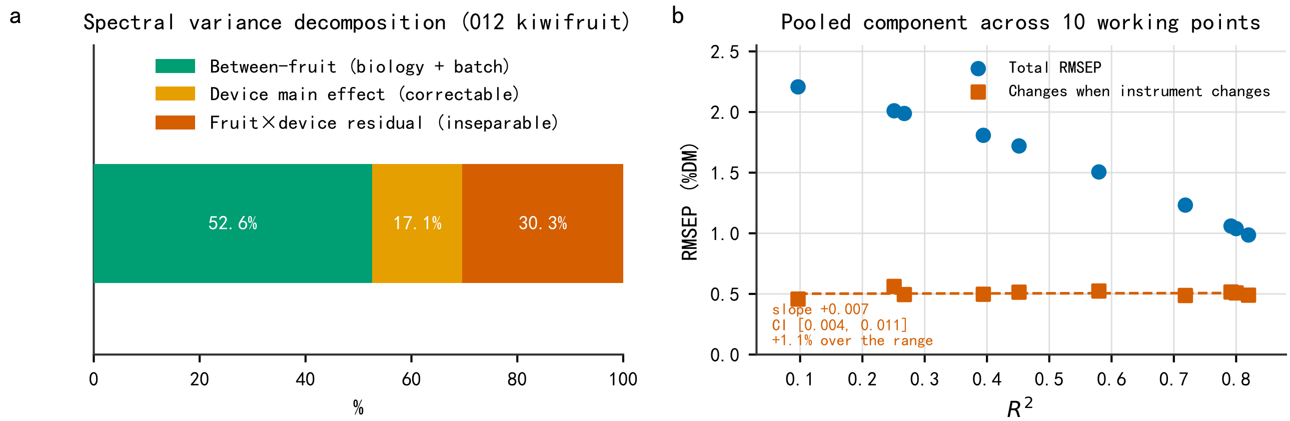

Figure 3: Between-instrument variability on the measurement side. (a) Three-component decomposition of the SNV-corrected spectral variance. (b) Total RMSEP (blue circles) and the pooled component that varies across devices (orange squares, being device×sample interaction together with scan/repositioning noise, which this design cannot separate) at ten pre-specified operating points, against R^2 on the abscissa. The dashed orange line is the linear regression of that component on R^2 and the band is its seed-clustered bootstrap 95% interval.

of nothing, and the size of the drop alone cannot distinguish the two. Supplementary S4 gives the full analysis, the terminal-date decomposition and Fig. S1.

3.5. A design table for the number of reference replicates

Inverting Eq. (2) answers how many independent replicate determinations the reference value must have for the attainable R^2 ceiling to reach a target $R^2_{\max}$:

$$m \geq \frac{\sigma_f^2}{\sigma_a^2} \cdot \frac{R^2_{\max}}{1 - R^2_{\max}}. \tag{4}$$

Equation (4) is the Spearman–Brown prophecy formula solved for the number of replicates [24, 25], as Supplementary S1 shows term for term. What this section adds is empirical and structural rather than algebraic: a σ_f^2/σ_a^2 measured on near-infrared calibration material, the target reliability read as an attainable R^2 ceiling, and the inversion written as a point prediction carrying no free parameters, which §3.6 tests against measurement at thirteen replicate levels.

Substituting the apple value $\sigma_f^2/\sigma_a^2 = 1.0965$ gives the design table: $R^2_{\max} = 0.7$ requires $m \geq 3$; 0.8 requires $m \geq 5$; 0.9 requires $m \geq 10$; 0.95 requires $m \geq 21$. The ratio already differs appreciably between the three origin groups of this collection (§3.1), so practical use should first estimate σ_f^2/σ_a^2 on a small subset of the target batch; batch membership is not recorded for this material, so no claim is made about how the ratio changes with batch. Equation (4) gives a structural lower bound, not a sufficient condition, because retraining a model on noisy reference values incurs a further loss the formula does not account for (§3.6). A margin above the bound is therefore advisable: the operational rule is to discount the clean baseline R^2_{clean} by 1% before inverting, a discount factor that is not the same quantity as the relative deviations reported there.

3.6. A direct test of the ceiling and the design rule

Sections 3.1–3.5 all rest on the apple data, whose spectral side performs poorly under the PLS protocols evaluated here, so this section meets the objection that a ceiling never approached by any working model might be a property that is never triggered. The answer is a semi-synthetic experiment whose criteria were archived before it was run and whose four design targets were fixed before the corresponding replicate levels had been measured. This is the second version of that pre-registration; the first is kept verbatim rather than deleted, and Supplementary S5 records what it set wrongly and that no threshold was changed in correcting it.

The analysis moves to the kiwifruit data, whose spectral signal is strong enough (calibration R^2 of 0.827 for DM and 0.861 for SSC), and injects reference noise into the labels: for each fruit the original label is replaced by the mean of m virtual replicate determinations, adding independent noise of variance σ_f^2/m. The kiwifruit label variance σ_y^2 is split into a between- and a within-fruit part in the ratio $\sigma_f^2/\sigma_a^2 = 1.0965$ measured on apples, and the within-fruit part is injected, $\sigma_f^2 = \left[1.0965/(1 + 1.0965)\right] \sigma_y^2 = 0.5230\, \sigma_y^2$, exactly the 52.30% within-fruit share of the apple data ($= 1 - \text{ICC}$); the split is stated rather than the ratio alone because the alternative reading $\sigma_f^2 = 1.0965\, \sigma_y^2$ differs by a factor of 2.1 and does not reproduce the required replicate counts of the blinded test below. The model is retrained on the noisy labels, the component count also being selected using them, which corresponds to the real situation in which the experimenter only ever holds noisy reference values. Thirteen levels of m are used $(1, 2, \ldots, 10, 14, 20, \infty)$ with the Formal five seeds, under the fruit-grouped repeated holdout that produces the values 0.827 and 0.861; that is not the 5-fold grouped cross-validation giving $R^2 = 0.819$ in §3.3, and the two must not be mixed.

Under this setup the model can perceive the entire clean label (variance σ_y^2) and only the injected σ_f^2/m is imperceptible to it, so the structural ceiling of Eq. (2) becomes $\sigma_y^2/(\sigma_y^2+\sigma_f^2/m)$ and the parameter-free prediction is

$$R^2_{\text{pred}}(m) = \frac{R^2_{\text{clean}}\,\sigma_y^2}{\sigma_y^2 + \sigma_f^2/m}. \tag{5}$$

At none of the 13 levels of m does the measurement cross the structural ceiling (Fig. 4). That is an arithmetic consequence of the clean baselines rather than an identity: the pre-registered tolerance $|R^2_{\text{obs}}/R^2_{\text{pred}} - 1| \leq 15\%$ is narrower than the factor R^2_{clean} by which R^2_{pred} already sits below the ceiling. The observation is consistent with the declared injected variance but does not by itself establish it, since staying below the ceiling is compatible with more than one injected magnitude; what the injection realises rests on the implementation and the pre-registered parameters, as Supplementary S5 sets out. The median relative deviation between measurement and Eq. (5), over the twelve finite-noise levels, is -0.9% (DM) and -0.3% (SSC). More notable is the shape: uniformly negative and shrinking monotonically as m grows (DM from -1.67% to -0.67%, SSC from -1.59% to -0.12%), which agrees with the mechanistic expectation that the less noise is injected, the less of it is fitted away during retraining, and which was not obtained by any fitting.

A direct test of the design rule. Four blinded targets were specified in advance, the m at which each lies having no measured result at the time of specification, inverted for the required replicate count and only then measured. What is inverted is Eq. (5) rather than Eq. (4), since the model here perceives the entire clean label and that baseline itself falls short of 1; Supplementary S5 gives the inverted expression, the four targets and their measured values. Two targets were met and two fell marginally short, all shortfalls within ± 0.008. This is the same phenomenon as the negative deviation above: Eq. (4) accounts only for the structural attenuation caused by reference noise, not for the additional loss incurred when the model is retrained on noisy labels. The rule is therefore restated accurately as a lower bound on the required number of replicates, with an advisory margin, namely discount R^2_{clean} by 1% before inverting. That margin changes the required m only for the two targets that previously fell short: one rises from $m = 7$ to $m = 8$, where the measured value is 0.8062, turning a shortfall into a pass, and this is the one occasion on which the margin rule is directly verified by measurement; the other rises from $m = 9$ to $m = 11$, a level outside the grid, so no measured value exists, no interpolation is performed, and the neighbouring levels merely bracket it. The 1% figure is accordingly an empirical margin induced from these four points rather than a universal constant, and it must be recalibrated on other data.

The demonstration is not equivalent to validation on the apple data. What it establishes is narrower: when the spectral signal is strong enough, reference noise limits performance in the predicted way. It cannot conversely explain the weak apple performance, and scaling the apple variance ratio onto kiwifruit gives the demonstration a noise magnitude with an empirical basis rather than asserting that the kiwifruit reference determinations carry variability of that size. This section and the attenuation-ratio test of §3.2 bracket the same proposition from opposite sides: §3.2 uses real replicate reference determinations but a weak model ($R^2 \approx 0.06$–0.12), so its limitation is power, not validity, whereas here the model is strong but the replicates are synthetic and the injected noise is i.i.d. by construction, so the most fragile premise of Eq. (2) is supplied by us rather than tested. The weaknesses are orthogonal and the argument requires both sides. The failure mode most likely in practice is excluded by neither: the tissue sampled destructively may be the same tissue that the effective optical sampling depth probes, which would make reference noise correlated with what the model can perceive, and ruling that out needs a dataset carrying both real replicate reference determinations and a working spectral model (§4.3).

4. Discussion

4.1. Interpretation of the existing literature

This paper does not claim that the high R^2 values reported in the literature are wrong. The split-leakage magnitude measured here, 1.41–1.46×, does not explain the gap to typical published values, and the apple data used here also perform poorly on the spectral side (§4.3), so this work cannot judge anyone else's model performance. The claim made is weaker and more robust: when the reference value is a single local determination, the ceiling of Eq. (2) holds for any predictor that can perceive only whole-fruit information, provided the mean-independence condition of §2.2 holds. That condition is what makes the statement model-free rather than linear-only, so the ceiling is a structural constraint independent of regressor, preprocessing and hyperparameters, and it must be stated precisely. If spectra and reference determinations are paired on the same face and local chemical differences do change the local spectrum, a model can in principle predict part of the biological content of e_{ij}, and Eq. (2) becomes a conditional ceiling for whole-fruit-level predictors only.

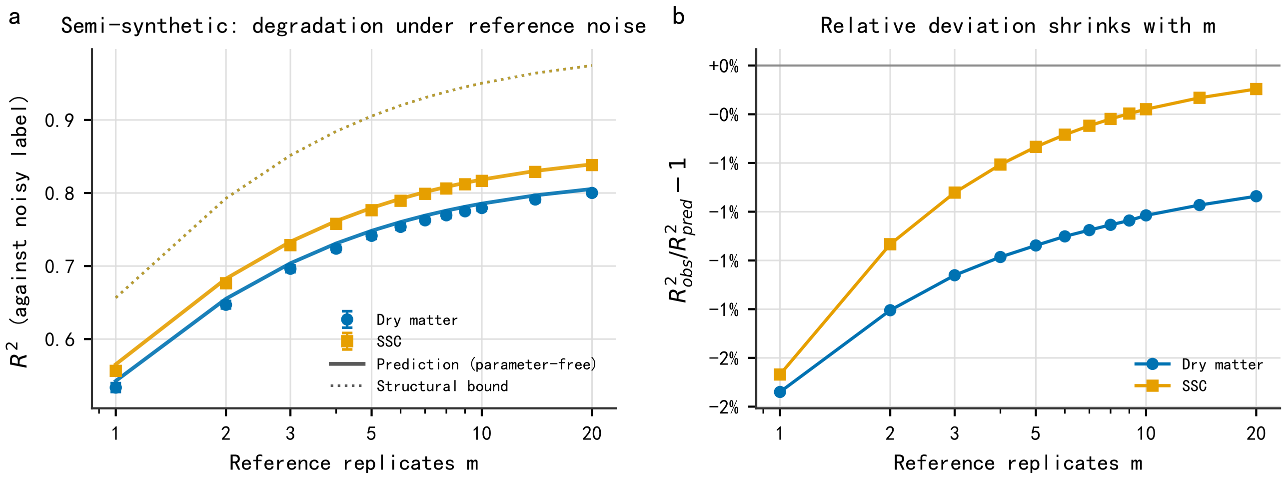

Figure 4: Semi-synthetic reference-noise degradation experiment. (a) Measured R^2 (points, error bars are seed-clustered bootstrap 95% CIs) against the prediction of Eq. (5) (solid) and the structural ceiling (dashed) as a function of the number of reference replicates m. (b) Shrinkage of the relative deviation $R^2_{\text{obs}}/R^2_{\text{pred}} - 1$ with m: uniformly negative and monotone, consistent with the mechanism that the extra retraining loss falls as the injected noise falls.

The apple data cannot distinguish the two cases, so the ceiling is asserted only under that condition.

The ceiling is not binding for every study. The apple data here measure $R^2 = 0.118$ on the whole-fruit mean target, so the ceiling to read it against is the five-face one, 0.820, not the single-face 0.477, and the measured value is far below either, so reference-value variability is not the binding constraint on that dataset's current performance. What does bind it is not established here: limited acquisition configuration is the hypothesis most consistent with the available diagnostics, but no experiment here varies that configuration and alternatives such as model class or parameter space were not individually excluded, so it is a candidate rather than a demonstrated cause. Whether a given study is bound depends on how far its performance sits from the ceiling, which the attenuation diagnostic can check directly on that study's own data.

4.2. Practical implications of the two components

On the reference side, the choice follows from the research target. If that target is whole-fruit quality, the reference value should be a multi-point mean and the required number of replicates can be fixed with the design table of §3.5; if it is local quality, spectra and reference determinations must be paired at the same location and the result must not be evaluated against a whole-fruit label. The attenuation diagnostic requires per-face replicate reference determinations on the same individual, so on one-fruit-one-label data it cannot be run and σ_f^2/σ_a^2 should be estimated first on a small subset. The practical yield is that the number of sampling faces per fruit can be fixed before an experiment starts, directing effort towards more reference replicates or a better model instead of towards tuning a model against a structurally unattainable target.

On the measurement side, an RMSEP reported on a single instrument omits the part of the error that appears only on cross-instrument deployment, and at the two operating points compared here the better the model, the larger that part's share of the total error (§3.3), so performance reports should give both an instrument-invariant part and a changes-with-instrument part. Calibration-transfer methods target the correctable device main effect [16, 17, 18, 3], but the fruit×device residual component cannot be separated from scan noise without technical replicates on the same device and is hard to remove by post-processing.

4.3. Limitations

The spectral side of the apple data performs poorly under the protocols evaluated here. Under fruit-grouped holdout the R^2 for whole-fruit mean SSC is only about 0.11, reproduced by two independently written implementations (0.118 and 0.109, overlapping intervals), which supports only insensitivity to implementation detail and not replication on independent data. Both are far below that dataset's five-face-mean ceiling of 0.820, so the conclusion is unaffected. Three diagnostics failed to support the candidate explanations: a row-wise random split gives only 0.169; mean saturation over the third overtone band of sugar (900–980 nm) is only 1.2% and restricting to non-saturated bands or to 900–1001 nm is worse than the full spectrum; and the same code and protocol give $R^2 = 0.827$ for DM and $R^2 = 0.861$ for SSC on the kiwifruit data. The single band most correlated with SSC is at 678 nm ($|r| = 0.154$), a colour–maturity proxy rather than direct sugar absorption. The reasonable attribution is twofold: reflectance geometry on intact fruit yields a surface-weighted sampling whose effective depth is typically shallow and varies with wavelength, geometry and fruit structure [41, 42], whereas the refractometer measures juice from destructively sampled flesh, and 391–1001 nm covers only the very weak third overtone. Per-diagnostic figures are in Supplementary S5.

The source of the within-fruit variability cannot be fully decomposed. With no technical replicates on the same sampling face, genuine spatial chemical heterogeneity cannot be separated from the variability of destructive sampling and preparation, and the term "within-fruit reference variability" is used throughout. Nothing in the practical conclusions is lost: provided face-level departures are approximately independent, which §3.2 tests, both sources decay as $1/m$ alike and Eqs. (2) and (4) are unchanged. Supplementary S5 gives that argument in full, with what the acquisition record does and does not contain.

The instrument component is a pooled term. With no technical replicates within a device, instrument$\times$ sample interaction and scan/repositioning noise are inseparable, so only the pooled term is reported and no claim is made about pure instrument repeatability. The decomposition is a sequential projection with neither REML nor rebalancing, and since the number of devices per fruit is not constant, unequal replication leaves the effect matrices no longer mutually orthogonal [43] and the two measurement shares order-dependent. The 1068–1137 nm columns are missing on all five devices, so the instrument budget covers only 402–1065 nm.

The two components come from different species; this is a division of labour, not mutual corroboration. They cannot validate each other, and the reason for placing them side by side is that no public dataset supplies both reference replicates and cross-instrument spectral replicates.

The three groups come from a single collection. All apple samples were purchased in a single 2025 acquisition from one wholesale market, and the record does not state whether the material constitutes a single batch; the nature and verification status of the origin labels are given in the note to Table 1. Since the ICC varies across the origin groups (0.335–0.615), the ceiling must be reported per collection. Reliability can vary with cultivar and season too, and modelling the variance components as heterogeneous is the established way to test that [44]; no such subgroup test was run, so the single ICC is an assumption rather than a finding.

Two preprocessing masks are computed before splitting. A paper about validation protocol should state where its own protocol falls short of the standard it argues for. The saturation mask behind the band-subset diagnostic and the missing-channel mask for the kiwifruit model are both built from the whole dataset and only then fed to the grouped holdouts. Both are target-blind, but under a strict no-clean-separation criterion [19] they remain decisions taken with held-out spectra in view, and a published route that removes the trade-off rather than bounding it [45] was not taken here. For the kiwifruit mask the question can be settled rather than bounded, because the mask removes 0 of 222 wavelength columns and a fold-local mask is therefore necessarily identical for every split. Separately, one exploratory script selects a preprocessing method by held-out R^2; no number from it enters this paper, and it is flagged because the released code contains it. The affected quantities and the direct check are in Supplementary S5.

5. Conclusions

The RMSEP reported in non-destructive fruit quality assessment contains two variance components that are not the model's doing. This paper gives a workflow for calibrating each of them separately and converts the result into a design rule usable in advance, $m \geq (\sigma_f^2/\sigma_a^2) \cdot R^2_{\max}/(1 - R^2_{\max})$. On the reference side, within-individual variability puts the attainable R^2 ceiling for apple SSC at 0.477 under a single-face reference value; across three origin groups bought in a single acquisition from one wholesale market that ceiling varies between 0.335 and 0.615, so it must be reported per collection rather than used as a species constant. On the measurement side, the pooled device$\times$sample and scan/repositioning component corresponds to about 0.49 %DM of RMSEP in kiwifruit dry-matter prediction. That absolute quantity is approximately invariant between the two model operating points compared here, the slope of its regression on R^2 having an interval that excludes zero while the fitted trend spans only 1.1% of the component's own magnitude, so under the strong model it accounts for 24.6% of the mean squared prediction error (two-point regime; 24.7% under the ten-point scan, §3.3).

The variance-component model underpinning these conclusions yields four groups totalling six parameter-free point predictions, whose measured values are all close to theory: the label-only baseline, for instance, has theoretical value 0.3462 against a measured 0.3456. No formal pass threshold was set for any of them except exchangeability across faces. On that basis we deliver two tools, a design table for the number of reference replicates and an attenuation-ratio diagnostic that any laboratory holding per-face (or per-point) replicate reference determinations can run on its own data. Both presuppose a leakage-free validation protocol, so we also specify the candidate grouping levels and the procedure for auditing them, delivering that audit procedure rather than a fixed nesting: the individual and the acquisition-day level are examined as parallel candidates, and whether either is necessary must be settled on the data at hand. In the apple data the individual level was confirmed necessary, whereas an apparently deeper acquisition-day-level leakage in the time-series data was withdrawn after a control test that attributes the drop mainly to a target derived from the grouping variable; one storage-duration combination still meets the tightened criterion, so a small genuine same-day shared-condition effect cannot be excluded.

Because the ceiling was never approached on the apple data, we ran a pre-registered semi-synthetic experiment (§3.6) on the kiwifruit data, injecting reference noise scaled by the variance ratio measured on apples and retraining. The degradation trajectory agrees with the parameter-free prediction at all 13 replicate levels (median signed relative deviations of −0.9% and −0.3%, always below the structural ceiling). That experiment also exposed a real boundary of the design rule: it accounts only for structural attenuation, not for the additional loss of retraining on noisy reference values, so it should be used as a lower bound. The 1% empirical margin proposed on that basis changes the requirement for only two of the four blinded targets, one directly verified and the other only bracketed, so it is an empirical value rather than a universal constant. The two variance components come from two datasets in a relation of division of labour, not mutual corroboration; a public dataset that provides replicate reference determinations and cross-instrument replicate scans at once does not yet exist, and how to calibrate both components on the same batch deserves further study.

Data and code availability

The apple data are unpublished primary research material held by Xinjiang Agricultural University. The institution's data policy does not permit public release, so they are not deposited in a public repository; they may be requested from the corresponding author by email, and are shared at the institution's discretion under that policy. Every quantity derived from them is released in full, as set out below. The kiwifruit data are public and were used exactly as deposited: Kiwifruit NIR, figshare, v1, CC BY 4.0, doi:10.6084/m9.figshare.21747983.v1 [28], first described by Wohlers et al. [29]. The third dataset is the public DeepHS-Fruit collection introduced by Varga et al. [27]; its 2023 five-species release was used as distributed, from https://github.com/cogsys-tuebingen/deephs_fruit.

All analysis scripts, intermediate result workbooks, per-run records, figures and the pre-registered statistical protocol are released at https://github.com/2004lryan/020nir-reference-error-budget, together with a claim-by-claim map, so every statistic reported here is checkable without access to the raw data. The intermediate products of the cross-architecture replay were not retained, so that replay can only be repeated from the scripts and seeds, not rechecked quantity by quantity.

CRediT authorship contribution statement

Panlin Li: Methodology, Software, Formal analysis, Visualization, Writing – original draft. Yuchang Li: Investigation, Data curation, Validation. Yutong Feng: Investigation, Data curation. Longjie Li: Resources, Investigation. Ya Liu: Resources, Investigation. Hua Huang: Conceptualization, Supervision, Project administration, Funding acquisition, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Funding

This work was supported by the Xinjiang Uygur Autonomous Region Natural Science Foundation, General Project (grant number pending), and by the 2026 Xinjiang Agricultural University College Students' Innovation Project (grant numbers S202610758116, DXSCX2026349, DXSCX2026350). The grant number of the Natural Science Foundation project is to be supplemented once the public announcement period closes. The funders had no involvement in study design, in the collection, analysis and interpretation of data, in the writing of the report, or in the decision to submit the article for publication.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Anthropic Claude in order to assist with language polishing and code review. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

References

[1] K.B. Walsh, J. Blasco, M. Zude-Sasse, X. Sun, Visible-NIR 'point' spectroscopy in postharvest fruit and vegetable assessment: The science behind three decades of commercial use, Postharvest Biol. Technol. 168 (2020) 111246. doi:10.1016/j.postharvbio.2020.111246.

[2] D. Zhang, Y. Xu, W. Huang, X. Tian, Y. Xia, L. Xu, S. Fan, Nondestructive measurement of soluble solids content in apple using near infrared hyperspectral imaging coupled with wavelength selection algorithm, Infrared Phys. Technol. 98 (2019) 297–304. doi:10.1016/j.infrared.2019.03.026.

[3] L. Li, W. Huang, Z. Wang, S. Liu, X. He, S. Fan, Calibration transfer between developed portable Vis/NIR devices for detection of soluble solids contents in apple, Postharvest Biol. Technol. 183 (2022) 111720. doi:10.1016/j.postharvbio.2021.111720.

[4] A.M. Rady, D.E. Guyer, N.J. Watson, Near-infrared Spectroscopy and Hyperspectral Imaging for Sugar Content Evaluation in Potatoes over Multiple Growing Seasons, Food Anal. Methods 14 (2021) 581–595. doi:10.1007/s12161-020-01886-1.

[5] A. Pissard, E.J.N. Marques, P. Dardenne, M. Lateur, C. Pasquini, M.F. Pimentel, J.A. Fernández Pierna, V. Baeten, Evaluation of a handheld ultra-compact NIR spectrometer for rapid and non-destructive determination of apple fruit quality, Postharvest Biol. Technol. 172 (2021) 111375. doi:10.1016/j.postharvbio.2020.111375.

[6] J. Wieme, K. Mollazade, I. Malounas, M. Zude-Sasse, M. Zhao, A. Gowen, D. Argyropoulos, S. Fountas, J. Van Beek, Application of hyperspectral imaging systems and artificial intelligence for quality assessment of fruit, vegetables and mushrooms: A review, Biosyst. Eng. 222 (2022) 156–176. doi:10.1016/j.biosystemseng.2022.07.013.

[7] P. Mishra, D. Passos, F. Marini, J. Xu, J.M. Amigo, A.A. Gowen, J.J. Jansen, A. Biancolillo, J.M. Roger, D.N. Rutledge, A. Nordon, Deep learning for near-infrared spectral data modelling: Hypes and benefits, TrAC Trends Anal. Chem. 157 (2022) 116804. doi:10.1016/j.trac.2022.116804.

[8] C.A. Garillos-Manliguez, J.Y. Chiang, Multimodal Deep Learning and Visible-Light and Hyperspectral Imaging for Fruit Maturity Estimation, Sensors 21 (2021) 1288. doi:10.3390/s21041288.

[9] F. Braga, M. Panteghini, The utility of measurement uncertainty in medical laboratories, Clin. Chem. Lab. Med. 58 (2020) 1407–1413. doi:10.1515/cclm-2019-1336.

[10] F. Braga, M. Panteghini, Performance specifications for measurement uncertainty of common biochemical measurands according to Milan models, Clin. Chem. Lab. Med. 59 (2021) 1362–1368. doi:10.1515/cclm-2021-0170.

[11] H. van Kippersluis, P. Biroli, R. Dias Pereira, T.J. Galama, S. von Hinke, S.F.W. Meddens, D. Muslimova, E.A.W. Slob, R. de Vlaming, C.A. Rietveld, Overcoming attenuation bias in regressions using polygenic indices, Nat. Commun. 14 (2023) 4473. doi:10.1038/s41467-023-40069-4.

[12] Q. Zhang, Q. Wang, Correcting for measurement error under meta-analysis of z-transformed correlations, Br. J. Math. Stat. Psychol. 77 (2024) 261–288. doi:10.1111/bmsp.12328.

[13] J. Ezenarro, D. Schorn-García, M. Plans, O. Busto, R. Boqué, Quantification of spectral measurement errors to guide preprocessing method selection: A case study on cannabinoid prediction across multiple NIR instruments, Anal. Chim. Acta 1343 (2025) 343705. doi:10.1016/j.aca.2025.343705.

[14] J. Ezenarro, J. Riu, H.J. Ahmed, O. Busto, B. Giussani, R. Boqué, Measurement errors and implications for preprocessing in miniaturised near-infrared spectrometers: Classification of sweet and bitter almonds as a case of study, Talanta 276 (2024) 126271. doi:10.1016/j.talanta.2024.126271.

[15] J. Riu, G. Gorla, M. Monti, M. Campeny, J. Quesada, B. Giussani, Measurement replicates under the lens: evaluating NIR spectroscopic error quality through multivariate error analysis, Anal. Chim. Acta 1419 (2026) 345885. doi:10.1016/j.aca.2026.345885.

[16] P. Mishra, R. Nikzad-Langerodi, F. Marini, J.M. Roger, A. Biancolillo, D.N. Rutledge, S. Lohumi, Are standard sample measurements still needed to transfer multivariate calibration models between near-infrared spectrometers? The answer is not always, TrAC Trends Anal. Chem. 143 (2021) 116331. doi:10.1016/j.trac.2021.116331.

[17] P. Mishra, D. Passos, Deep calibration transfer: Transferring deep learning models between infrared spectroscopy instruments, Infrared Phys. Technol. 117 (2021) 103863. doi:10.1016/j.infrared.2021.103863.

[18] J. Yang, J. Li, J. Hu, W. Yang, X. Zhang, J. Xu, Y. Zhang, X. Luo, K.C. Ting, T. Lin, Y. Ying, An interpretable deep learning approach for calibration transfer among multiple near-infrared instruments, Comput. Electron. Agric. 192 (2022) 106584. doi:10.1016/j.compag.2021.106584.

[19] S. Kapoor, A. Narayanan, Leakage and the reproducibility crisis in machine-learning-based science, Patterns 4 (2023) 100804. doi:10.1016/j.patter.2023.100804.

[20] M. Rosenblatt, L. Tejavibulya, R. Jiang, S. Noble, D. Scheinost, Data leakage inflates prediction performance in connectome-based machine learning models, Nat. Commun. 15 (2024) 1829. doi:10.1038/s41467-024-46150-w.

[21] P. Király, G. Tóth, Being Aware of Data Leakage and Cross-Validation Scaling in Chemometric Model Validation, J. Chemom. 39 (2025) e70026. doi:10.1002/cem.70026.

[22] J. Correa, H. Tavakoli, S. Vogel, R. Gebbers, Overfitting due to data leakage in soil sensor calibration: Examples from lab-based and in-situ soil NIR spectroscopy, Comput. Electron. Agric. 239 (2025) 110920. doi:10.1016/j.compag.2025.110920.

[23] A. Vabalas, E. Gowen, E. Poliakoff, A.J. Casson, Machine learning algorithm validation with a limited sample size, PLoS One 14 (2019) e0224365. doi:10.1371/journal.pone.0224365.

[24] C. Spearman, Correlation calculated from faulty data, Br. J. Psychol. 3 (1910) 271–295. doi:10.1111/j.2044-8295.1910.tb00206.x.

[25] W. Brown, Some experimental results in the correlation of mental abilities, Br. J. Psychol. 3 (1910) 296–322. doi:10.1111/j.2044-8295.1910.tb00207.x.

[26] T.J. Smith, T.J.B. Kline, A. Kline, GeneralizIT: A Python solution for generalizability theory computations, SoftwareX 31 (2025) 102235. doi:10.1016/j.softx.2025.102235.

[27] L.A. Varga, J. Makowski, A. Zell, Measuring the Ripeness of Fruit with Hyperspectral Imaging and Deep Learning, in: 2021 International Joint Conference on Neural Networks (IJCNN), IEEE, 2021, pp. 1–8. doi:10.1109/IJCNN52387.2021.9533728.

[28] [dataset] M. Wohlers, Kiwifruit NIR, figshare, v1, 2025. doi:10.6084/m9.figshare.21747983.v1.

[29] M. Wohlers, A. McGlone, E. Frank, G. Holmes, Augmenting NIR Spectra in deep regression to improve calibration, Chemom. Intell. Lab. Syst. 240 (2023) 104924. doi:10.1016/j.chemolab.2023.104924.

[30] D. Liljequist, B. Elfving, K. Skavberg Roaldsen, Intraclass correlation – A discussion and demonstration of basic features, PLoS One 14 (2019) e0219854. doi:10.1371/journal.pone.0219854.

[31] D. Schiano-Lomoriello, K.J. Hoffer, I. Abicca, G. Savini, Repeatability of automated measurements by a new anterior segment optical coherence tomographer and biometer and agreement with standard devices, Sci. Rep. 11 (2021) 983. doi:10.1038/s41598-020-79674-4.

[32] O. Gerke, Reporting Standards for a Bland–Altman Agreement Analysis: A Review of Methodological Reviews, Diagnostics 10 (2020) 334. doi:10.3390/diagnostics10050334.

[33] R.J. Janse, T. Hoekstra, K.J. Jager, C. Zoccali, G. Tripepi, F.W. Dekker, M. van Diepen, Conducting correlation analysis: important limitations and pitfalls, Clin. Kidney J. 14 (2021) 2332–2337. doi:10.1093/ckj/sfab085.

[34] P. Mishra, D.N. Rutledge, J.-M. Roger, K. Wali, H.A. Khan, Chemometric pre-processing can negatively affect the performance of near-infrared spectroscopy models for fruit quality prediction, Talanta 229 (2021) 122303.

doi:10.1016/j.talanta.2021.122303.

[35] A.C. Burnett, J. Anderson, K.J. Davidson, K.S. Ely, J. Lamour, Q. Li, B.D. Morrison, D. Yang, A. Rogers, S.P. Serbin, A best-practice guide to predicting plant traits from leaf-level hyperspectral data using partial least squares regression, J. Exp. Bot. 72 (2021) 6175–6189. doi:10.1093/jxb/erab295.

[36] K. Meacham-Hensold, C.M. Montes, J. Wu, K. Guan, P. Fu, E.A. Ainsworth, T. Pederson, C.E. Moore, K.L. Brown, C. Raines, C.J. Bernacchi, High-throughput field phenotyping using hyperspectral reflectance and partial least squares regression (PLSR) reveals genetic modifications to photosynthetic capacity, Remote Sens. Environ. 231 (2019) 111176. doi:10.1016/j.rse.2019.04.029.

[37] H. Wada, K. Nakata, H. Nonami, R. Erra-Balsells, M. Tatsuki, Y. Hatakeyama, F. Tanaka, Direct evidence for dynamics of cell heterogeneity in watercored apples: turgor-associated metabolic modifications and within-fruit water potential gradient unveiled by single-cell analyses, Hortic. Res. 8 (2021) 187. doi:10.1038/s41438-021-00603-1.

[38] T.S. Madssen, A. Smilde, J. Camacho, A.H. Jarmund, J. Westerhuis, G.F. Giskeødegård, Statistical Validation of Multivariate Treatment Effects in Longitudinal Study Designs, J. Chemom. 39 (2025) e70044. doi:10.1002/cem.70044.

[39] D. Wilimitis, C.G. Walsh, Practical Considerations and Applied Examples of Cross-Validation for Model Development and Evaluation in Health Care: Tutorial, JMIR AI 2 (2023) e49023. doi:10.2196/49023.

[40] M.J. Lewis, A. Spiliopoulou, K. Goldmann, C. Pitzalis, P. McKeigue, M.R. Barnes, nestedcv: an R package for fast implementation of nested cross-validation with embedded feature selection designed for transcriptomics and high-dimensional data, Bioinform. Adv. 3 (2023) vbad048. doi:10.1093/bioadv/vbad048.

[41] A. Rodríguez-Ortega, N. Aleixos, J. Blasco, F. Albert, S. Munera, Study of light penetration depth of a Vis-NIR hyperspectral imaging system for the assessment of fruit quality. A case study in persimmon fruit, J. Food Eng. 358 (2023) 111673. doi:10.1016/j.jfoodeng.2023.111673.

[42] Y. Huang, J. Xiong, Z. Li, D. Hu, Y. Sun, H. Jin, H. Zhang, H. Fang, Recent Advances in Light Penetration Depth for Postharvest Quality Evaluation of Fruits and Vegetables, Foods 13 (2024) 2688. doi:10.3390/foods13172688.

[43] M. de Figueiredo, S. Rudaz, J. Boccard, A unified strategy to rebalance multifactorial designs with unequal group sizes: application to analysis of variance multiblock orthogonal partial least squares, Anal. Chim. Acta 1263 (2023) 341284. doi:10.1016/j.aca.2023.341284.

[44] P. Martinková, F. Bartoš, M. Brabec, Assessing Inter-rater Reliability With Heterogeneous Variance Components Models: Flexible Approach Accounting for Contextual Variables, J. Educ. Behav. Stat. 48 (2023) 349–383. doi:10.3102/10769986221150517.

[45] O.-C.G. Engstrøm, M.H. Jensen, Fast Partition-Based Cross-Validation With Centering and Scaling for $X^\top X$ and $X^\top Y$, J. Chemom. 39 (2025) e70008. doi:10.1002/cem.70008.